MIT-ADT UNIVERSITY
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# MIT SCHOOL OF COMPUTING

## REAL-TIME SIGN LANGUAGE TRANSLATION

## COMPUTER VISION-POWERED ACCESSIBLE COMMUNICATION FOR THE DEAF AND HARD-OF-HEARING COMMUNITY

### Problem Statement

Deaf and hard-of-hearing individuals face severe daily communication barriers because most people do not understand sign language. Existing translation options are often inaccessible—human interpreters are costly and unavailable for spontaneous interactions, while existing tech solutions require bulky sensor gloves or expensive depth cameras.

### Research Question

How can computer vision and deep learning enable real-time, accessible sign language translation using only a standard webcam, without specialized hardware or wearable sensors?

### Scope & Feasibility

✓ Translates dynamic sign gestures into real-time text using a standard mobile camera.

✓ Run fully on-device without internet.

✓ Enables easy addition of custom sign vocabulary via built-in data recording.

✓ Runs lightweight MediaPipe hand keypoint tracking efficiently on consumer CPUs.

✓Provide extensible dataset recording pipeline.

✓ Classifies compact coordinate vectors for fast, real-time AI inference.

### Key Features


Core technical capabilities of the translation system:

- 🖐 Real-Time Hand Skeleton Tracking: Extracts 21 3D hand keypoints per frame using a standard webcam feed.
- 📐 Wrist-Relative Normalization: Zero-centers and scales landmarks relative to the wrist, making gestures distance- and camera-angle invariant.
- 🎬 Dynamic Motion Recognition: Processes 30-frame sequence buffers (~1 second of motion) to translate fluid, moving gestures.
- 🔒 On-Device & Private Execution: Runs all computer vision tracking and neural network inference locally without sending video streams over the internet.
- 💻 Hardware Agnostic: Operates smoothly on standard 720p webcams with zero specialized hardware or wearable sensors required.
- 📦 Dataset Recording Pipeline: Built-in tool to capture and export 30-frame numpy array sequences for easy training of custom sign vocabulary.

### Proposed Solution

A lightweight computer vision pipeline using MediaPipe for hand tracking feeds normalized keypoint sequences into an LSTM neural network trained on ASL gestures, producing real-time text translation output on any standard device.

### System Architecture

Webcam Input → MediaPipe Hand Tracking → 21 Keypoint Extraction → Wrist-Relative Normalization → 30-Frame Sequence Buffer → LSTM Classifier → Text Output Display
All processing occurs locally on-device, ensuring user privacy and low latency response times under 100ms.

### Team

**Group ID -** SY205
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